Example 2.24

Multiply $\sqrt[3]{40}$ and $\sqrt[3]{16}$.

Solution

$$\sqrt[3]{40} \times \sqrt[3]{16} = \left(\sqrt[3]{2\times2\times2\times5}\right) \times \left(\sqrt[3]{2\times2\times2\times2}\right)$$
$$= \left(2\times\sqrt[3]{5}\right)\times\left(2\times\sqrt[3]{2}\right) = 4\times\left(\sqrt[3]{2}\times\sqrt[3]{5}\right) = 4\times\sqrt[3]{2\times5}$$
$$= 4\sqrt[3]{10}$$

Example 2.25

Compute and give the answer in the simplest form: $2\sqrt{72}\times5\sqrt{32}\times3\sqrt{50}$

Solution

$$2\sqrt{72}\times5\sqrt{32}\times3\sqrt{50} = \left(2\times6\sqrt{2}\right)\times\left(5\times4\sqrt{2}\right)\times\left(3\times5\sqrt{2}\right)$$
$$= 2\times5\times3\times6\times4\times5\times\sqrt{2}\times\sqrt{2}\times\sqrt{2}$$
$$= 3600\times2\sqrt{2}$$
$$= 7200\sqrt{2}$$

Let us simplify:

$\sqrt{72} = \sqrt{36\times2} = 6\sqrt{2}$

$\sqrt{32} = \sqrt{16\times2} = 4\sqrt{2}$

$\sqrt{50} = \sqrt{25\times2} = 5\sqrt{2}$

Example 2.26

Divide $\sqrt[9]{8}$ by $\sqrt[6]{6}$.

Solution

$$\frac{\sqrt[9]{8}}{\sqrt[6]{6}} = \frac{8^{\frac{1}{9}}}{6^{\frac{1}{6}}}$$ (Note that 18 is the LCM of 6 and 9)

$$= \frac{8^{\frac{2}{18}}}{6^{\frac{3}{18}}}$$ (How?)

$$= \left(\frac{8^2}{6^3}\right)^{\frac{1}{18}}$$ (How ?) $$= \left(\frac{8\times8}{6\times6\times6}\right)^{\frac{1}{18}}$$

$$= \left(\frac{8}{27}\right)^{\frac{1}{18}} = \left[\left(\frac{2}{3}\right)^3\right]^{\frac{1}{18}} = \left(\frac{2}{3}\right)^{\frac{1}{6}} = \sqrt[6]{\frac{2}{3}}$$



Activity - 1

Is it interesting to see this pattern ?

$$\sqrt{4\frac{4}{15}} = 4\sqrt{\frac{4}{15}} \quad \text{and} \quad \sqrt{5\frac{5}{24}} = 5\sqrt{\frac{5}{24}}$$

Verify it. Can you frame 4 such new surds?

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